MSc Final Project · 2026

Predicting Persuasive Comments on r/ChangeMyView

Length, Rhetorical Strategies, and Cross-Corpus Validation

Dikla Vilge

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How Does ChangeMyView Work?



Original View

OP posts an opinion





Reply

Another user challenges it





Delta Awarded

If persuaded,
OP awards a Δ

Can we predict which replies receive a Δ?

Three Research Questions

01 Can we predict persuasion from text?

And how much does comment length matter?

02 Do rhetorical strategies add predictive value?

Beyond length and other textual features?

03 Which strategies matter most?

And do the findings hold across datasets?

Our Dataset: Predicting Delta Awards

500 discussion posts from r/ChangeMyView

3,971
Comments

641
Delta-awarded comments

16.1%
Positive class

Task: Predict Δ vs. no Δ from comment features

Primary metric: PR-AUC



Initial Approach: A Rich Feature Pipeline

229 Engineered features

Reply Features

Linguistic, stylistic, and sentiment signals

Original Post Features

Characteristics of the original viewpoint

Reply–Post Relations

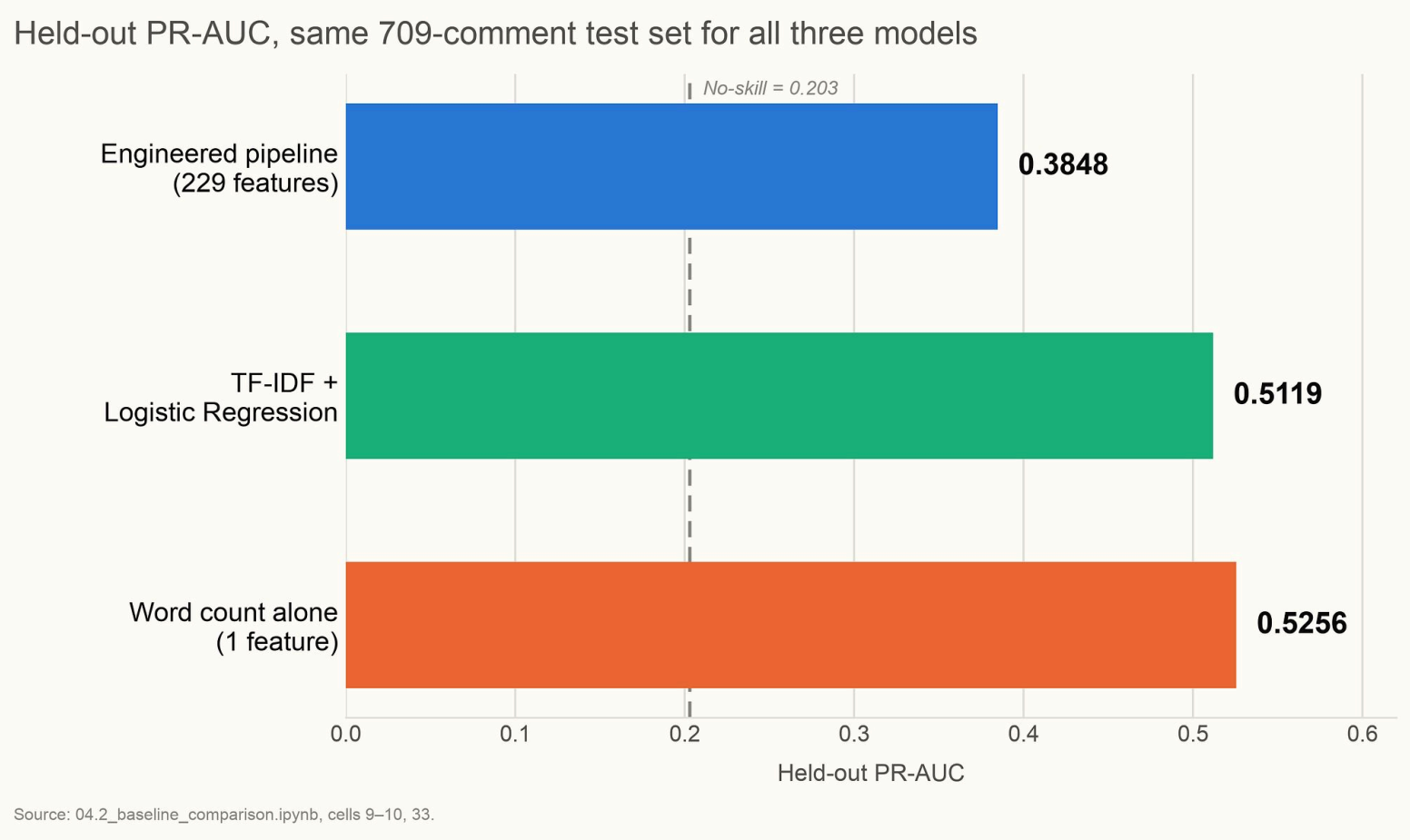
Similarity and contextual relationships

Combined features matrix

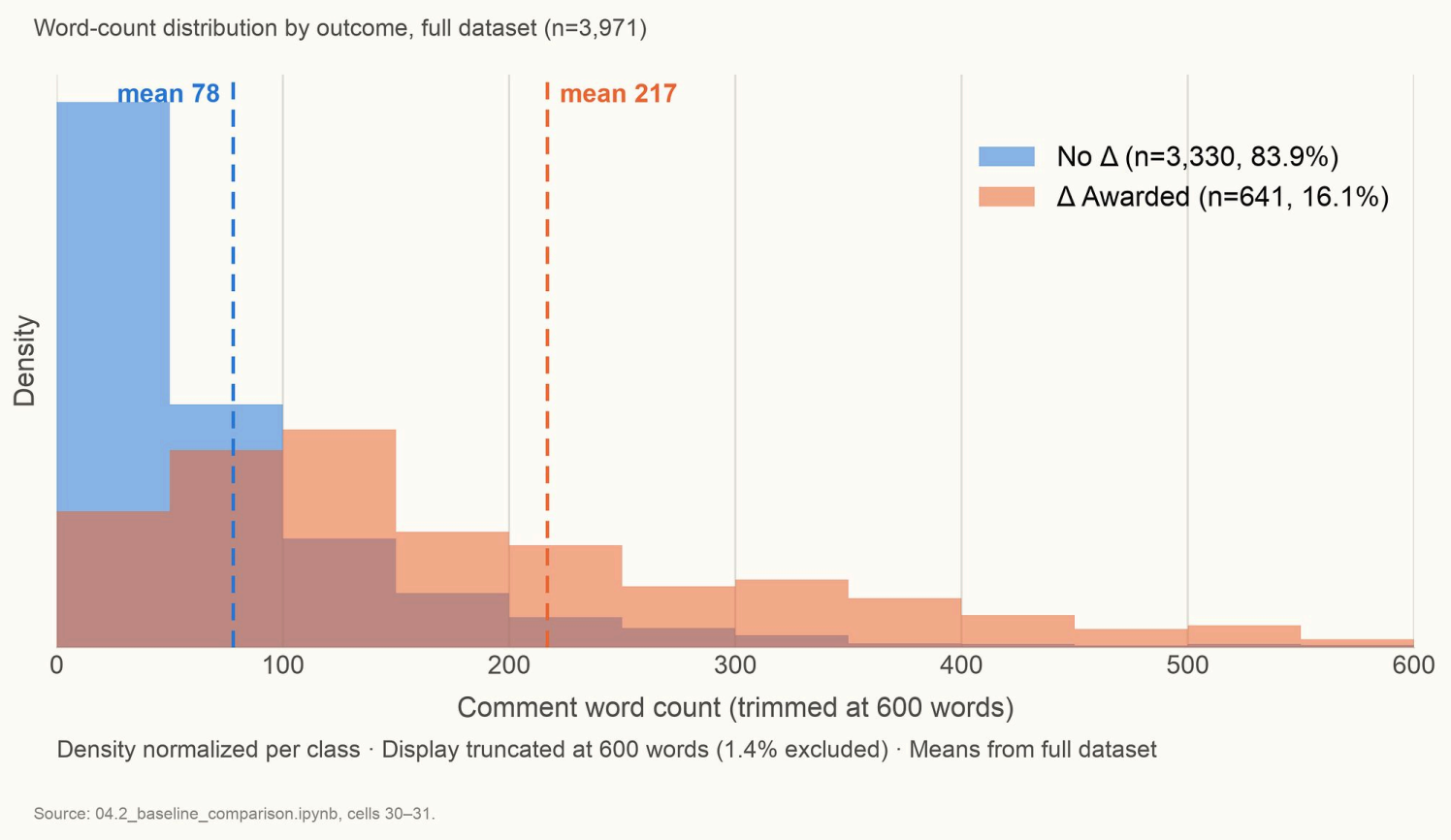
↓

Model Selection → XGBoost

Pivot: Length Beats the Engineered Pipeline



Delta-Awarded Comments Are Nearly 3× Longer



No Δ

78

mean words

Δ Awarded

217

mean words

- Within-thread check: Longer in 76.5% of mixed-outcome threads

Eight Rhetorical Strategies

Gemini scored each strategy independently on a 1–5 scale · 2,753 comments

Justification

Giving reasons

Concession

Acknowledging a point

Evidence

Supporting with facts

Refutation

Countering an argument

Personal Narrative

Sharing experience

Call to Action

Urging action

Simplification

Making ideas clearer

Question Asking

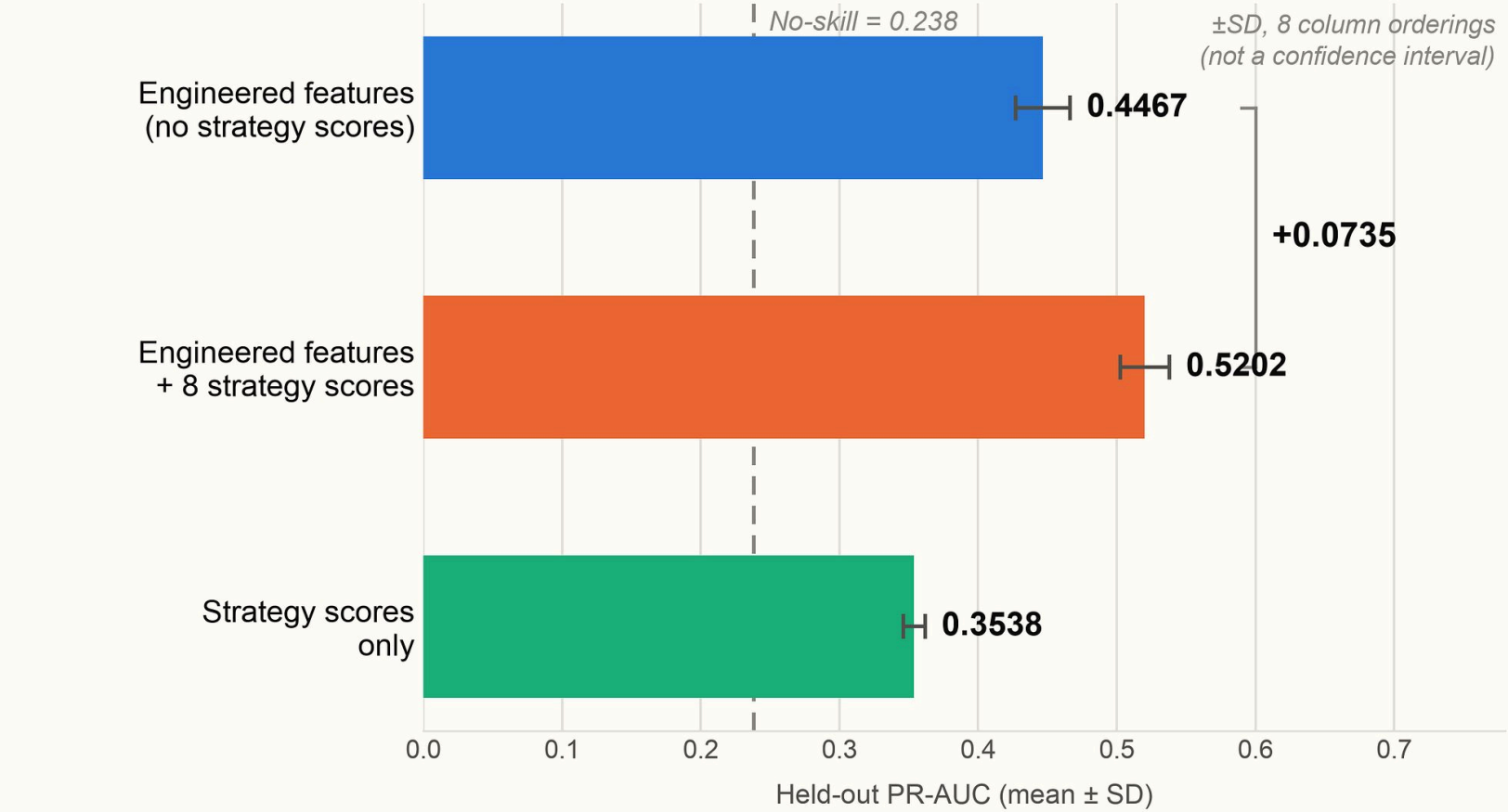
Using questions

Do Rhetorical Strategies Improve Prediction?

+0.0735

PR-AUC improvement

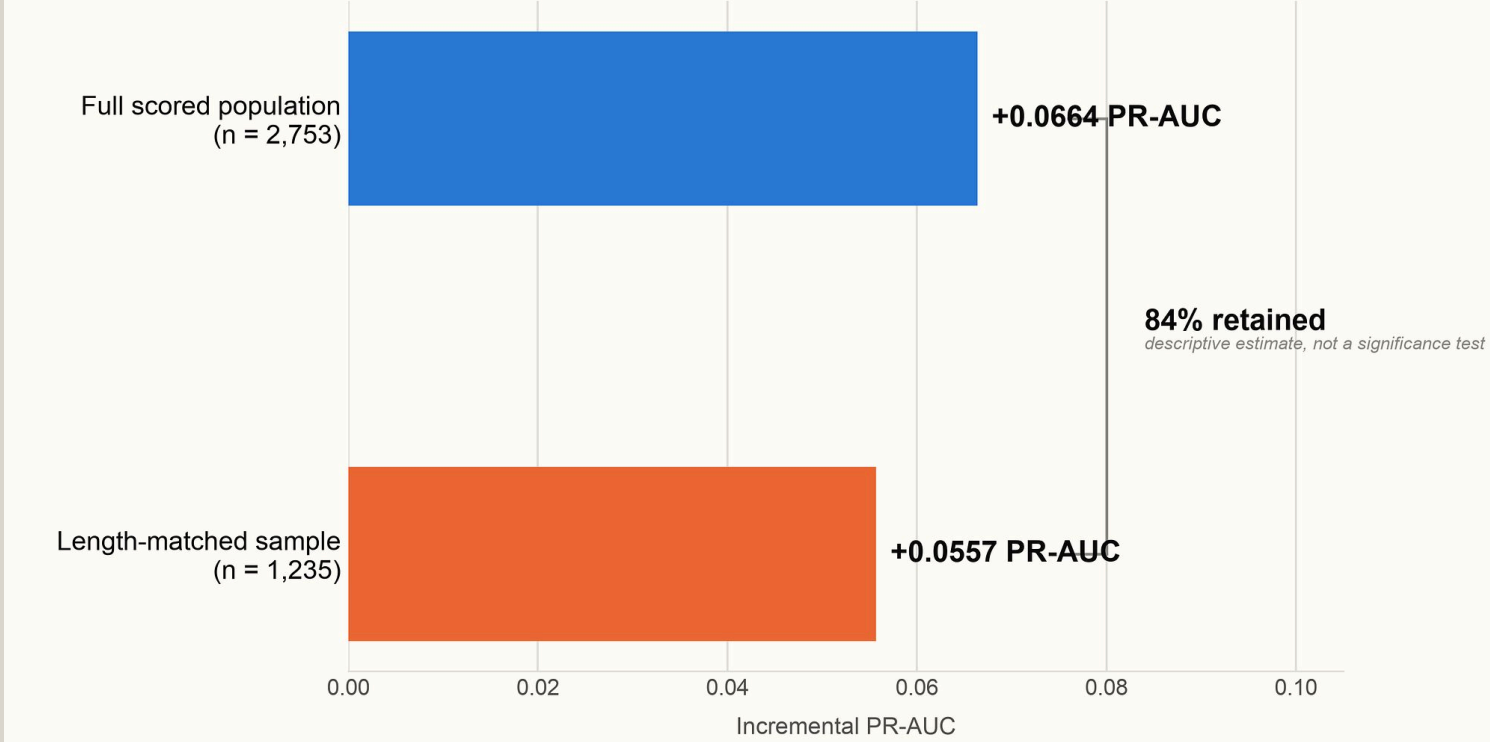
With rhetorical strategy scores



Scored subset, n=2,753 (test: 592 comments)

Source: 06_strategy_ablation.ipynb, cell 17 (headline_df). Column-order sensitivity documented in cells 11–14.

Does the Gain Survive Length Control?



Matched by word-count decile bin, not exact-length pairs. A separate, non-comparable sample exists in 04.3 (not used here). Source: 06_strategy_ablation.ipynb, cells 32, 35.

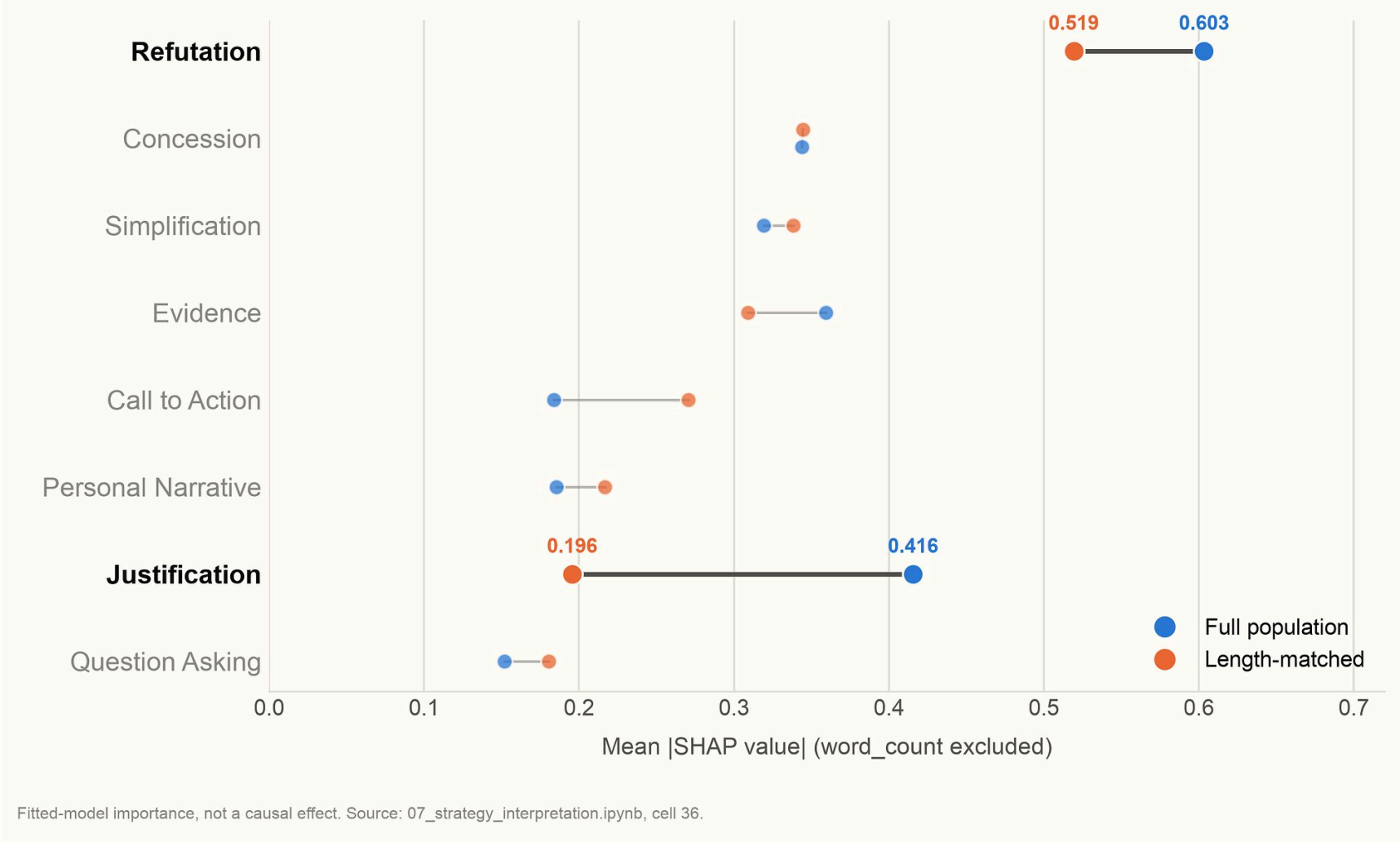
84%

of the predictive gain retained

After length matching

Descriptive comparison across different samples; not proof of independence from length.

Refutation Remains the Top Strategy After Length Matching



Fitted-model importance, not a causal effect. Source: 07_strategy_interpretation.ipynb, cell 36.

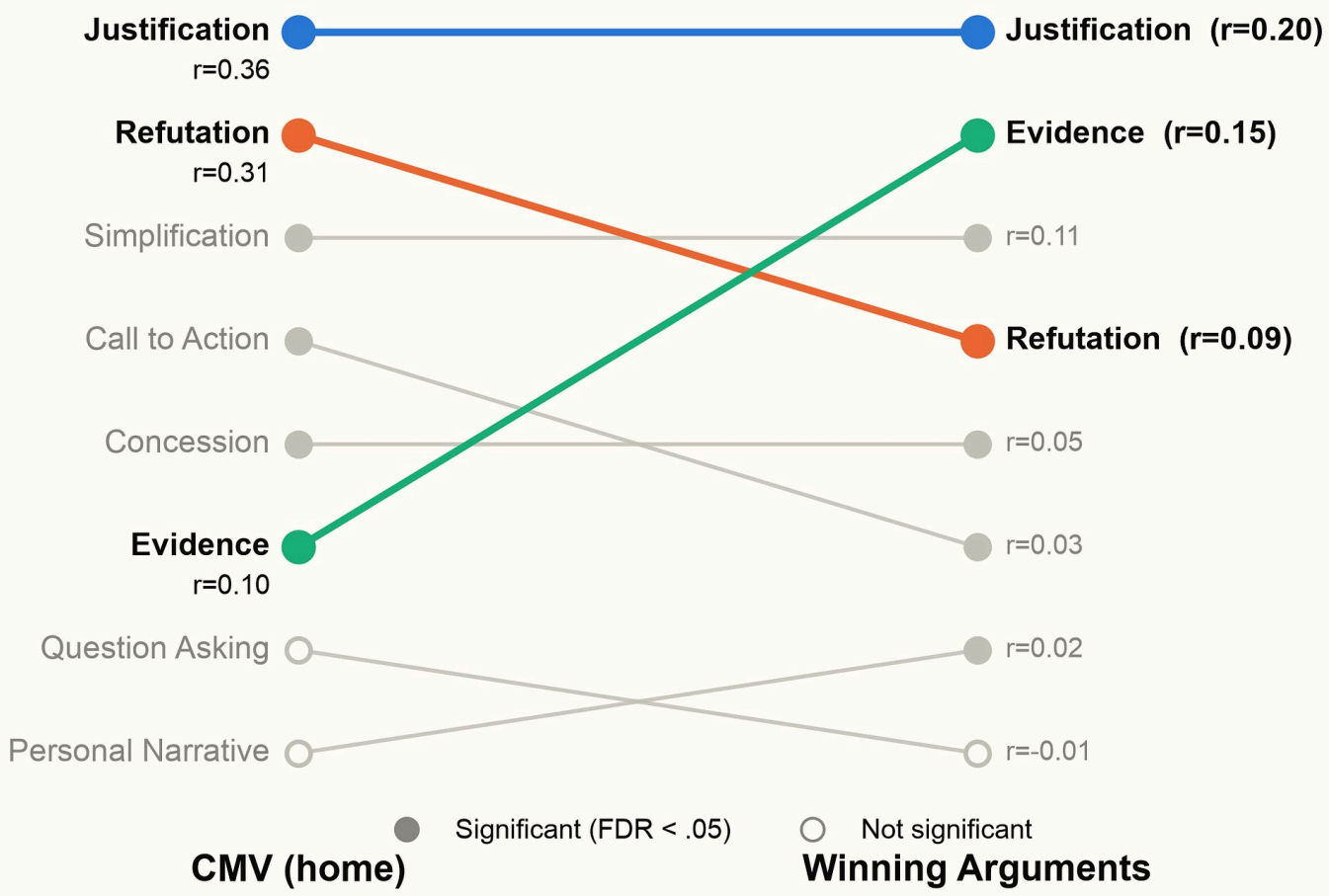
Justification's importance drops substantially

Important caveat

Refutation and justification showed the weakest agreement with human ratings.

Rhetorical Strategy Rankings Shift Across Datasets

Univariate association with outcomes · Rank-biserial effect size



Source: 07_strategy_interpretation.ipynb cell 39 (home), 08_wa_external_validation.ipynb cell 33 (WA).

Evidence: 6th → 2nd

Rises in Winning Arguments

Refutation: 2nd → 4th

Falls in Winning Arguments

Rhetorical Strategies Outperform Length Alone on WA

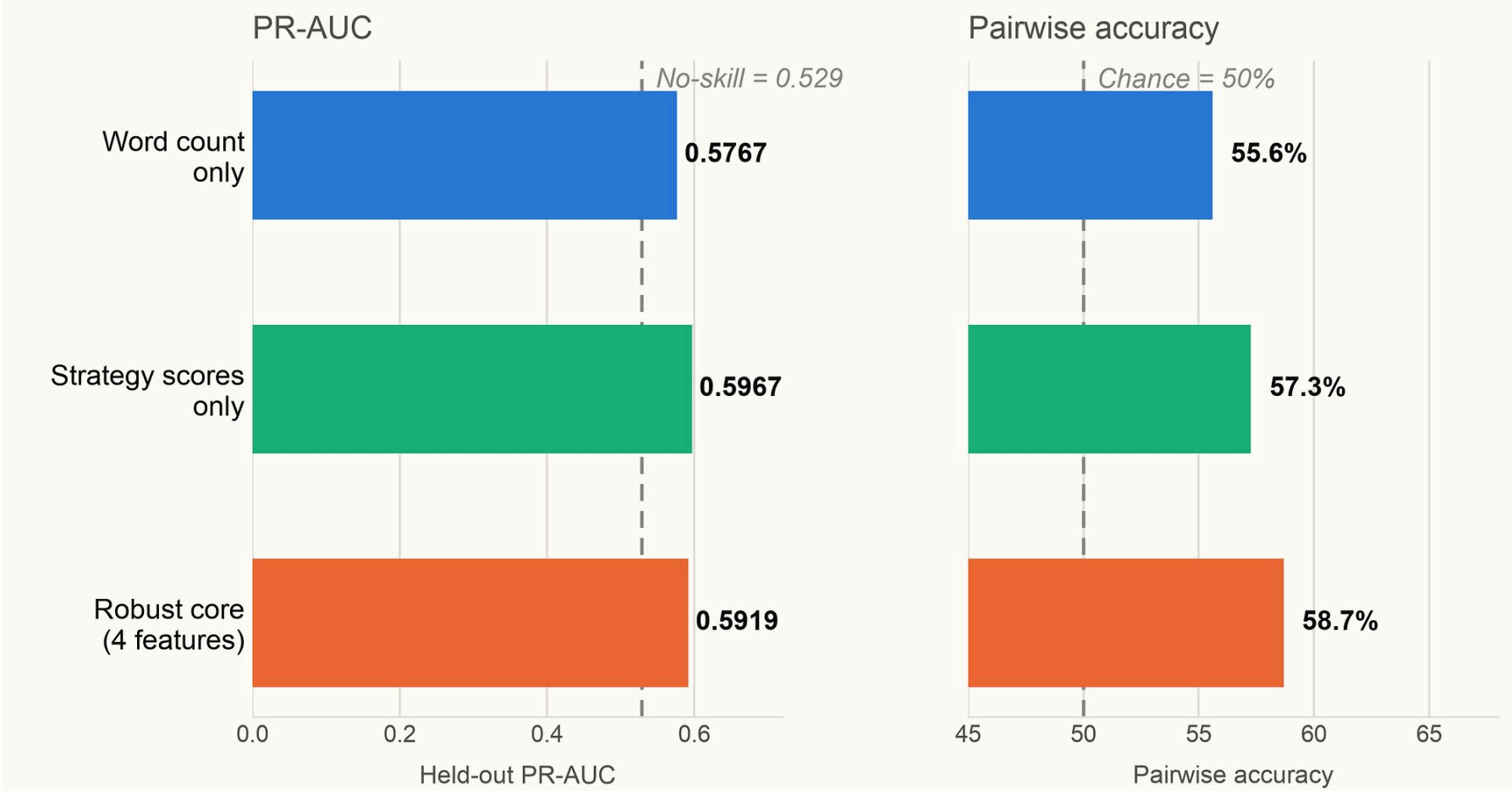
Retrained on WA

Evaluated on WA held-out data

Different metrics, different winners

Strategies lead on PR-AUC

4-feature core leads on pairwise accuracy



Source: 08_wa_external_validation.ipynb, cells 20, 38–39.

Conclusions: Beyond Length, Context Matters

Length Is a Powerful Baseline

Word count alone outperformed the 229-feature pipeline on home dataset.

Rhetorical Strategies Add Predictive Value

Strategy-score gains largely persisted after length matching.

Strategy Rankings Depend on Context

Evidence rose and refutation fell in Winning Arguments.

Contribution: Systematic comparison of engineered features, rhetorical strategies, and cross-corpus patterns.

Limitations: Observational study; small human-validation sample; no comparable length-controlled ablation on WA.

Thank you!

Questions?

Could a 9-Feature Model Replace the Full Pipeline?

Full model · 237 features

PR-AUC: 0.5202 ± 0.0177

Compact model · 9 features

PR-AUC: 0.5379 ± 0.0109

8 strategy scores + word count

Does Model Architecture Matter?

Random Forest

0.5432 ± 0.0065

XGBoost

0.5380 ± 0.0108

Values: mean ± SD across feature-column orderings (not confidence intervals).

Small difference relative to observed variation; **no clear architectural winner.**

Why XGBoost? Kept for consistency with the main experimental pipeline

How Stable Are the PR-AUC Results?

0.0422

PR-AUC variation from changing feature-column order alone



Same data



Same model architecture



Different feature-column order

Why?

Correlated features can have nearly equal split gains, making XGBoost sensitive to feature ordering.

Our response

Report mean ± SD across multiple column orderings for the affected CMV feature comparisons.

Why Didn't We Report the Best WA Score?

Pre-specified robust core · 4 features

PR-AUC: 0.5919

Pairwise accuracy: 58.7%

Reported in main results

Best-found variant · 8 features

PR-AUC: 0.5937

Pairwise accuracy: 59.1%

Strategy set without Call to Action

Why report the 4-feature core?

The 4-feature core was specified in advance.

Reporting the best-found WA variant as the headline result could overstate performance.